6 Receivers to Detect Digital Pulse

Objective

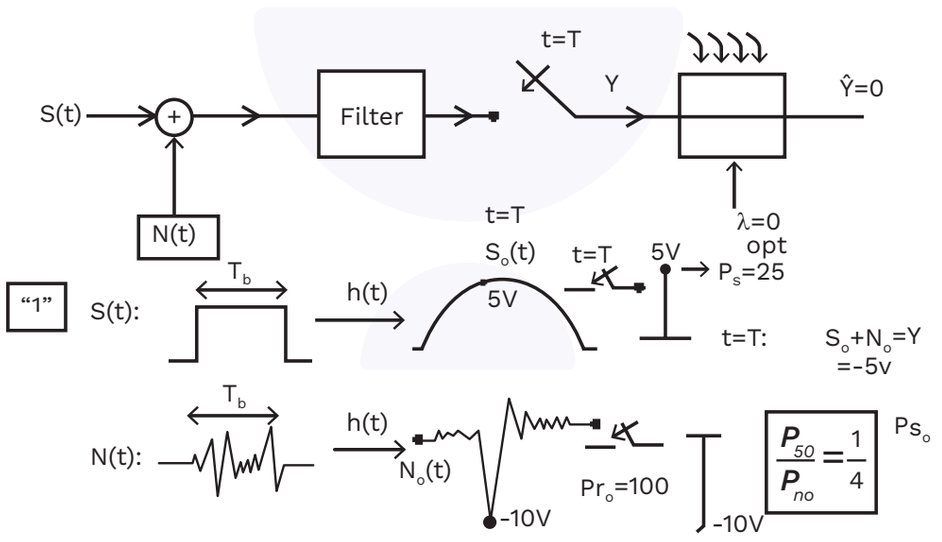
After completing this chapter, user will be able to understand:

- About the digital receiver.
- Matched filter.
- Reception of digital signals and effect of noise.

Introduction

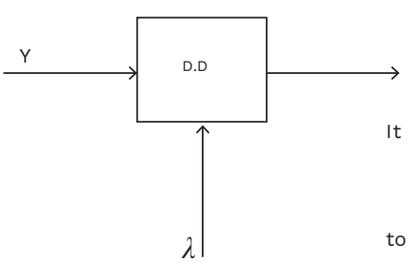
- The purpose of this chapter is to consider the various systems for the transmission of digital data and their relative performance.
- We will make the assumption that source symbols occur with equal probability.
- Digital communication systems may be coherent or non-coherent and synchronous or asynchronous.
- The primary measure of system performance is the probability of error.
- Synchronous detection in a white Gaussian noise requires a correlation or matched filter detector to give the minimum probability of error for fixed signal and noise conditions.

DIGITAL RECEIVER



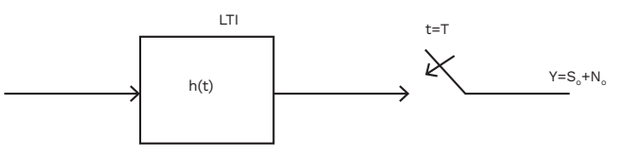
Blocks Responsible for Reduction of P_e:

1. Decision device:



It adjusts λ to λ opt to minimise Pe.

2.



It should be designed in such a way that $\frac{P_{So}}{P_{No}}$ or SNR should be maximum at the out-put of the fitter at the sampling instant.

Designing of Optimum Filter

Purpose: To maximise SNR at t =1

Assumptions: Let channel noise is white noise

→ Let channel noise is independent of signal

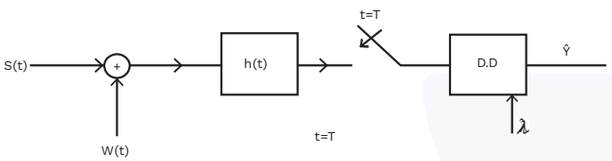
N(t) = W(t)

$$E[w(t)] = 0$$

$$Sw(f) = \frac{N_o}{2}$$

Let, the filter is to be designed for a single pulse:

$$S(t) \rightarrow 0 \le t \le T_b$$

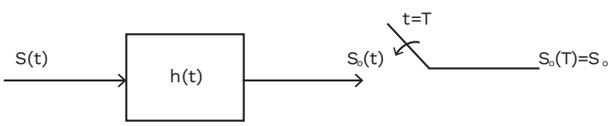


Note: For proper designing, duration of S(t) should match with sampling instance $T = T_b$.

Signal Analysis

S(t): Energy signal/pulse

$$E_s = \int_{-\infty}^{\infty} |\, s(t) \,|^2 \, dt = \int_{-\infty}^{\infty} |\, S(f) \,|^2 \, df$$

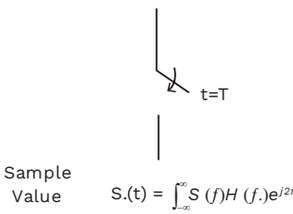


$S_0(T)=S(t)*h(t)$

$S_0(f)=S(f)H(f)$

IFT $S_0(t) = \int_{-\infty}^{\infty} S_0(f) e^{j2\pi ft} df$

$S_0(t) = \int_{-\infty}^{\infty} S(f) H(f) e^{j2\pi ft}$



Signal Power at Sampling Instant

$$P_{s.} = I_{s.}(T)^2 = I \int_{-\infty}^{\infty} S(f)H(f)e^{L2\pi fT} dfI^2$$

Schwartz Inequality

Schwartz Inequality

$$\int_{-\infty}^{\infty} X_1(f) X_2(f) df \,^2 \le \int_{-\infty}^{\infty} X_1(f) \,^2 df \int_{-\infty}^{\infty} X_1(f) \,^2 df$$

$$\int_{-\infty}^{\infty} X_1(f) X_2(f) df \,^2_{\max} = \int_{-\infty}^{\infty} X_1(f) \,^2 df \int_{-\infty}^{\infty} X_2(f) \,^2 df$$

Only When $X_2(f) = X_1^*(f)$

Maximum signal power at the sampling instant

$$P_{so} = \left| \int_{-\infty}^{\infty} \underbrace{S(f)}_{X_1(f)} \underbrace{H(f) e^{j2\pi ft}}_{X_2(f)} \right|^2 df \,|\, AB \,| = |\, A \,|\, B \,|$$

$$(P_{so})_{\max} = \int_{-\infty}^{\infty} |\, S(f) \,|^2 \, df \int_{-\infty}^{\infty} |\, H(f) e^{j2\pi fT} \,|^2 df$$

only when

$$X_1(f) = X_2^*(f)$$

$$(P_{so})_{\max} = \int_{-\infty}^{\infty} |\, S(f) \,|^2 \, df \int_{-\infty}^{\infty} |\, H(f) \,|^2 \, df$$

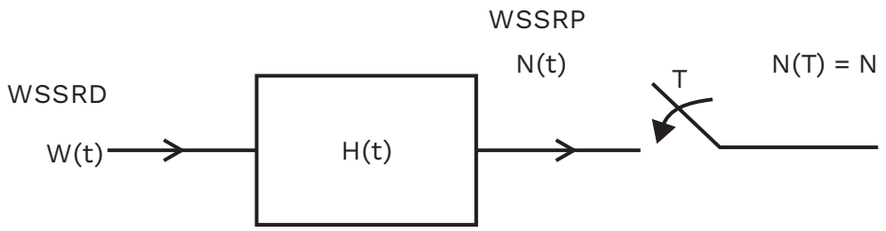
only when

If $S(f) = H^*(f) e^{-j2\pi fT}$

Then $\{Ps_o\}_{\max} = \int_{-\infty}^{\infty} |\, S(f) \,|^2 \, df \int_{-\infty}^{\infty} |\, H(f) \,|^2 \, df$

$$\{Ps_o\}_{\max} = E_{S(t)} \times E_{n(t)}$$

Noise Analysis



$$Sw(f) = \frac{N}{2}$$

$$P_N = E[N^2] = E[N^2(t)] = \int_{-\infty}^{\infty} |H(f)|^2 \times \frac{N_0}{2}\,df = \frac{N_0}{2} \times E_{n(t)}$$

P_N: noise power at sampling instant

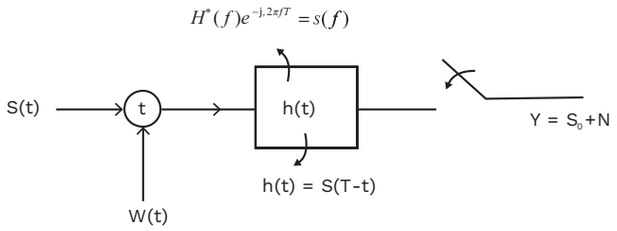
$\frac{N_0}{2}$: constant

$(SNR)_{max}$ at sampling instant t = T

$$\frac{(P_{S0})_{max}}{P_N} = \frac{E_{n(t)} \times E_{s(t)}}{\frac{N_o}{2} \times E_{n(t)}}$$

$$\boxed{\frac{(P_{s_o})_{max}}{P_N} = \frac{E_{S(t)}}{(N_o/2)}} = \boxed{(SNR)_{max} = \frac{E_{s(t)}}{(N_o/2)}}$$

Conclusion



1. $(SNR)_{max}$ at t = T $\rightarrow \boxed{\frac{\{P_{so}\}_{max}}{(P_n)} = \frac{E_{s(t)}}{(N_o/2)}}$
2. Above maximisation is obtained by designing the filter in a certain way $S(f) = H^*(f)e^{-j2\pi fT}$

Filter Design

$$S(f) = H^*(f)e^{-j2\pi fT}$$

$S(t)$: Real $\quad S(t) = S^*(t)$

$F.T$

$S(f) = S^*(-f)$

$S^*(f) = S(-f)$

$$H^*(f) = S(f)e^{j2\pi fT} \quad or \quad H(f) = \left[S(f)e^{j2\pi fT}\right]^*$$

$$S(t) \leftrightarrow S(f)$$

$$S(t+T) \leftrightarrow S(f)e^{+j2\pi fT}$$

$$S(-t+t) \leftrightarrow S(-f)e^{-j2\pi fT}$$

$$\boxed{h(t) = S(-t+T)} \leftrightarrow S^*(f)e^{-j2\pi fT} = \left[s(f)e^{-j2\pi fT}\right]^* = H(f)$$

Design Codⁿ

$$\boxed{H^*(f)e^{-L2\pi fT} = S(f)}$$

$$\boxed{h(t) = S(T-t)}$$

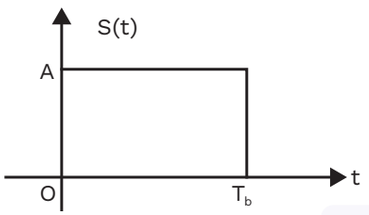
S(t) $\xrightarrow{\text{Fold}}$ S(-t) $\xrightarrow[\text{by T}]{\text{Right shift}}$ S(T-t) = h(t)

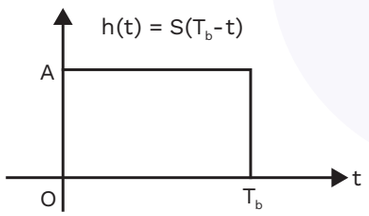
T: duration of s(t) = sampling instant

Example:

$$S(t)=\begin{cases}A & 0\le t\le T_b \\ 0 & elsecohere\end{cases} \qquad N(t)=W(t)$$

1. Unit impulse response of the optimum filter to maximise SNR at a sampling instant.
2. Output SNR at the sampling instant





$$(SNR)_{max}=\frac{P_S}{P_N}=\frac{E_{S(t)}}{N_{0/2}}=\frac{A^2T_b}{N_{0/2}}$$

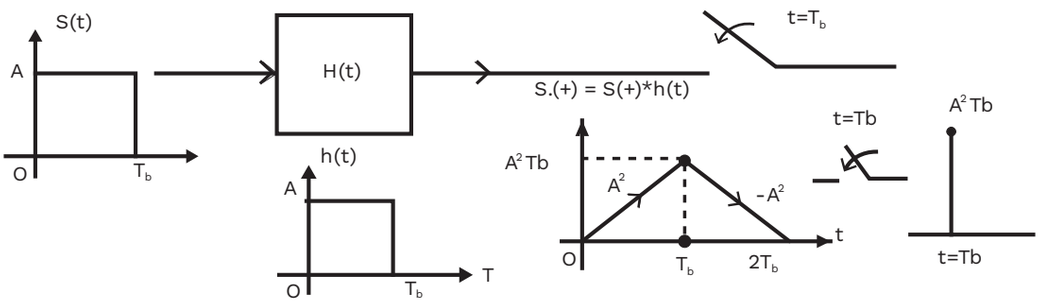
Matched Filter

1. It maximises the SNR at the output of filter sampling instant by adjusting its unit impulse response h(t).
2. It is also known as optimum Rx.
3. In matched filter receives channel noise is always white.

Properties of Matched Filters

Let the duration of input pulse S(t) is T, and the sampling instant is also T.

1. $h(t) = S(T-t)$
2. $s(t) = h(T-t)$
3. $E_{s(t)} = E_{h(t)}$
4. So(t) = S(t) * h(t)
   a) duration of S(t) and h(t) are same asT
   b) duration of So(t) is 2T
5. $|S_0(t)|_{max} = E_s(t) = E_n(t)$
6. $$\{SNR\}\ \frac{(PS_0)_{max}}{P_n}=\frac{E_{S(t)}}{N_0/2}=\frac{E_{n(t)}}{N_0/2}$$



h(t) = S(T_b-t)

S(t) : duration : T_b

h(t): duration : T_b

→ duration of h(t) will always with s(t)

optimum Filter → MATCHED FILTER

7) $S_1(t) \longrightarrow (SNR_1)_{max}\ at\ t=T \rightarrow \frac{Es_{1(t)}}{N_{0/2}}$

$S_2(t) \longrightarrow (SNR_2)_{max}\ at\ t=T \rightarrow \frac{Es_{2(t)}}{N_{0/2}}$

If

$E_{s1}(t) > E_{s2}(t)$

$S_1(t) \longrightarrow P_{e1}$

$S_2(t) \longrightarrow P_{e2}$

$P_{e2} > P_{e1}$

If $S_1(t)$, $S_2(t)$, $S_3(t)$, and $S_4(t)$ are Txed such that

$E_{s1}(t) > E_{s2}(t) > E_{s3}(t) > E_{s4}(t)$

$P_{e1} < P_{e2} < P_{e3} < P_{e4}$

$S_o(f) = S(f)\ H(f)$

$S_o(f) = S(f)\ S^*(f)e^{-j2\pi Ft}$

$S_o(f) = |S(f)|^2 e^{-j2p\,Ft}$

$S(t) \longleftrightarrow S(f) \longrightarrow |S(F)|^2 = G_s(F)$

Energy pulse — ESD of S(

$S_o(f) = G_s(f)e^{-j2\pi fT}$

$R_s(\tau) \overset{FT}{\longleftrightarrow} G_s(f)$

ACF of S(t) — ESD of S(t)

Inverse

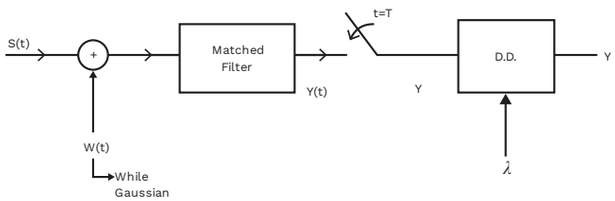
$S_o(t) = R_s(t-\tau)$

$(SNR)_{max}\ at\ t=T \rightarrow \frac{(Ps_o)_{max}}{P_N} = ?$

$(Ps_o)_{max} = E_{s(t)} \times E_{n(t)}$

$P_n = E_{n(t)} \times \frac{N_o}{2}$

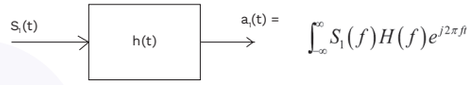
Matched filter based Rx

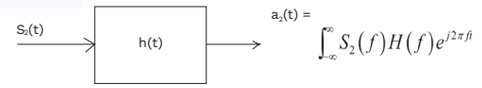


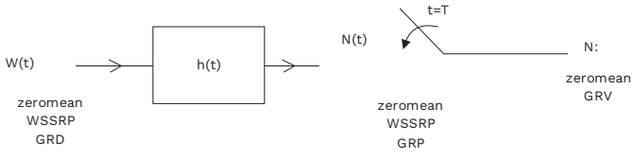
$$S(t)=\begin{cases} S_1(t):0\le t\le T_b & 1T_x \\ S_2(t):0\le t\le T_b & 0T_x \end{cases}$$

$$S(t)+W(t)=\begin{cases} S_1(t)+W(t):0\le t\le T_b & 1T_x \\ S_2(t)+W(t):0\le t\le T_b & 0T_x \end{cases}$$

$$S_o(t)+N(t)=\begin{cases} a_1(t)+N(t) \\ a_2(t)+N(t) \end{cases}$$







$$Y = S_o + N = \begin{cases} a_1 + N & 1\,Tx \\ a_2 + N & 0\,Tx \end{cases}$$

GRV — GRV — GRV

$$E(Y)=\begin{cases} a_1 & : \ 1T_x \\ a_2 & : \ oT_x \end{cases}$$

$$\sigma_y{}^2=\begin{cases} \sigma_N{}^2 & : \ 1T_x \\ \sigma_N{}^2 & : \ 0T_x \end{cases}$$

$$\boxed{\begin{aligned} \sigma_N{}^2 &= E\left[N^2\right]+\left(E\left[N\right]\right)^2 \\ \sigma_N{}^2 &= E\left[N^2\right]+E\left[N^{2(+)}\right]=\frac{N_o}{2}\int_{-\infty}^{\infty}|H(f)|^{2df} \end{aligned}}$$

$$P_e = P(oT_x)P\left(\frac{1}{oT_x}\right)+P(1T_x)P\left(\frac{o}{1T_x}\right)$$

$$\boxed{P\left(\frac{1}{OT_x}\right)=P[Y(OT_x)>\text{ก}\ ]=Q\left[\frac{\lambda-u_2}{\sigma_2}\right]=Q\left[\frac{\lambda-o_2}{\sigma_N}\right]}$$

$$\boxed{P\left(\frac{1}{1T_x}\right)=P[Y(2T_x)<]\ \ 1-Q\left[\frac{\lambda-u_1}{\sigma_2}\right]=1-Q\left[\frac{\lambda-a_1}{\sigma_N}\right]}$$

$$\boxed{\begin{aligned} &\rightarrow \sigma^2{}_1=\sigma^2{}_2=\sigma^2{}_N \\ &\rightarrow \text{If } P(oT_x)=P(1T_x)=1/2 \end{aligned}} \longrightarrow$$

$$\boxed{\lambda_{opt}=\frac{\mu_1+\mu_2}{2}=\frac{a_1+a_2}{2}\rightarrow P_e\,|_{\min}=Q\left[\frac{a_1-a_2}{2\sigma_N}\right]}$$

$$\left\{\begin{matrix} \text{due to } \lambda_{opt} \text{ or} \\ \text{Decision Devise} \end{matrix}\right\}$$

$$P_e|_{\min}=Q\left[\frac{a_1-a_2}{2\sigma_N}\right]=Q\left[\frac{x}{2}\right]$$

$$x=\frac{a_1-a_2}{\sigma_N}$$

$$\boxed{as\ x\uparrow\ Q\left[\frac{x}{2}\right]\downarrow}$$

So if x is maximised then P_e can be further reduced.

$$x=\frac{a_1-a}{\sigma_N}$$

#

Maximisation of $|x|^2$ leads to maximisation of x

$$|x|^2=\left|\frac{a_1-a_2}{\sigma_N}\right|^2=\frac{|\,a_1-a_2\,|^2}{|\,\sigma_N\,|^2}$$

$$a_1=a_1(T)=\int_{-\infty}^{\infty}s_1(f)H(f)e^{j2\pi fT}df$$

$$a_2=a_2(T)=\int_{-\infty}^{\infty}s_2(f)H(f)e^{j2\pi fT}df$$

$$\sigma^2{}_N=\frac{N_0}{2}\int_{-\infty}^{\infty}|H(f)|^2df$$

$$|x|^2=\frac{|\int_{-\infty}^{\infty}s_1(f)H(f)^{j2\pi ft}\,df-\int_{-\infty}^{\infty}s_2(f)H(f)e^{j2\pi ft}df\,|^2}{\frac{N_o}{2}\int_{-\infty}^{\infty}|\,H(f)\,|^2\,df}$$

$$|x|^2=\frac{\left|\int_{-\infty}^{\infty}(s_1(f)-s_2(f)e^{j2\pi fT}\frac{H(f)}{x_1(f)}\frac{H(f)}{x_2(f)}df\right|^2}{\frac{N_o}{2}\int_{-\infty}^{\infty}|\,H(f)\,|^2df}$$

If $X^*{}_1(f)=X_2(f)$

$$Then\ |x|^2{}_{\max}=\frac{\int_{-\infty}^{\infty}|\,s_1(f)-s_2(f)\,|^2\,df\int_{-\infty}^{\infty}|H(f)|^2df}{\frac{N_0}{2}\int_{-\infty}^{\infty}|\,H(f)\,|^2\,df}$$

$$x|^2{}_{\max}=\frac{\int_{-\infty}^{\infty}|\,s_1(f)-s_2(f)\,|^2\,df}{\left(N_o/2\right)}=\frac{E_d}{\left(N_o/2\right)}$$

$$P(t)=s_1(t)-s_2(t)\leftrightarrow P(f)=s_1(f)-s_2(f)$$

$$E_d=\int_{-\infty}^{\infty}|\,p^2(t)\,|\,dt=\int_{-\infty}^{\infty}|\,P(f)\,|^2\,df$$
$$=\int_{-\infty}^{\infty}|\,s_1(f)-s_2(f)\,|^2\,df$$

$$\boxed{x_{\max}=\sqrt{|\,x\,|^2{}_{\max}}=\sqrt{\frac{2E_d}{N_o}}}$$

$$P_e=P(oT_x)P\left(\frac{1}{oT_x}\right)+P(1T_x)P\left(\frac{o}{1T_x}\right)$$

(Channel noise white) +

$$\left(P(oT_x)=P(1T_x)=1/2\right)+\lambda_{opt}$$

$$P_e|_{min} = Q\left[\frac{a_1 - a_2}{2\sigma_N}\right] = Q\left[\frac{x}{2}\right]$$

↓ Matched filter

$$P_e|_{min|_{min}} = Q\left[\frac{x_{max}}{2}\right] = Q\left[\sqrt{\frac{E_d}{2N_o}}\right]$$

DESIGN :

$$\left[s_1(f)e^{j2\pi fT} - s_2(f)e^{j2\pi fT}\right]^* = H(f)$$

$$h(t) = s_1(T-t) - s_2(T-t)$$

NOTE: *'Design of matched filters'.*

1. $S(t) : O \le t \le T \rightarrow h(t) = S(T-t)$
2. $S_1(t) = O \le t \le T$

   $S_2(t) = O \le t \le T$ → h(t) = s1 (T-t)-S2 (T-t)

Formula of Pe:

AWGN

1) $P(oT_x) = P(1T_x) = 1/2$
2) Channel Noise: white
3) Filter: matched
4) λ_{opt}

$$P_e|_{|_{min}|_{min}} = Q\left[\sqrt{\frac{E_d}{2N_o}}\right]$$

$$E_d = \int_o^T |P^2(t)|^2 \, dt = \int_o^T |s_1(t) - s_2(t)^2| \, dt$$

Example: In a certain PCM systems, NRZ coding scheme is used to transmit bit 1 and bit 0 with equal probability through AWGN (Additive White Gaussian Noise) channel. If matched filter based receiver having MAP based decision making is used, calculate.

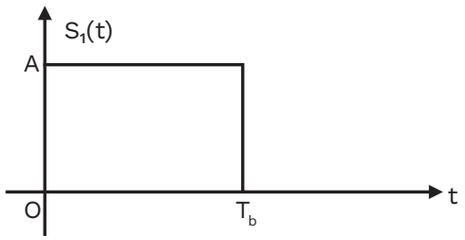
1. Impulse response of matched filter
2. P_e
3. P_e for AWGN channel
4. IID

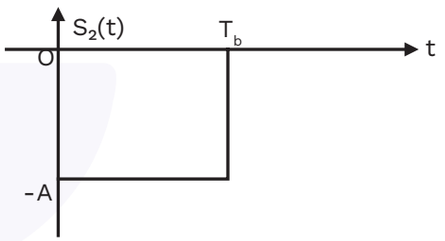
Solution:

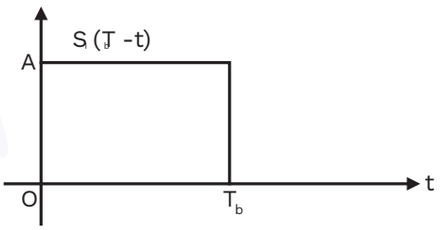
$P(oT_x) = P(1T_x) = 1/2$
Channel noise : white
Filter : matched
λ_{opt}

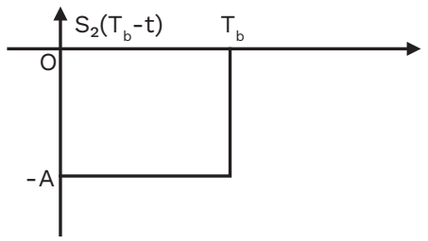
→ (ii) $P_e = Q\left[\sqrt{\frac{E_d}{2N_o}}\right]$

1. $h(t) = s_1(T-t) - S_2(T-t)$

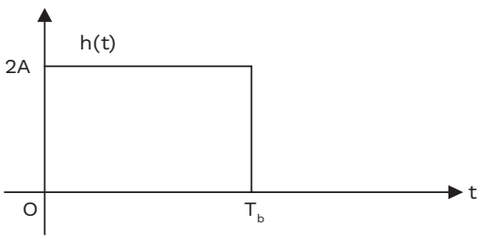








$h(t) = S_1(T_b - t) - S_2(T_b - t) = 2A \; O \le t \le T_b$



2. $P_e = Q\left[\sqrt{\frac{E_d}{2N_o}}\right]$

$E_d = \int_o^{T_b} |S_1(t) - S_2(t)|^2 \, dt = \int_o^{T_b} 4A^2 dt = 4A^2T_b$

$P_e = Q\left[\sqrt{\frac{2A^2T_b}{N_o}}\right]$

$(E_b)_{avg} = P(oT_x)E_{bo} + 1P(1T_x)E_{b1} = E_b$

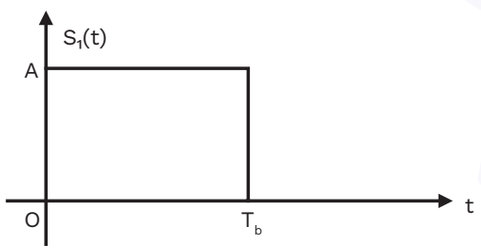
$\frac{1}{2} \times A^2T_b + \frac{1}{2} \times A^2T_b = E_b$

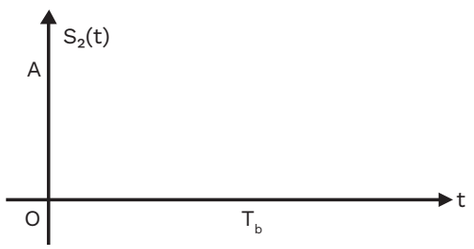
$E_b = A^2T_b$

$P_e = Q\left[\sqrt{\frac{2E_b}{N_o}}\right]$

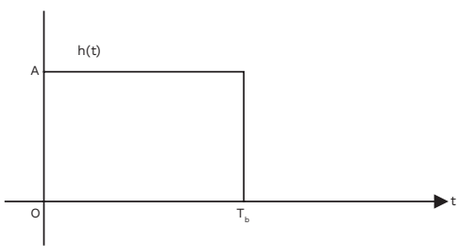
3. $P_e = Q\left[\sqrt{\left(\frac{2A^2T_b}{N_o}\right)^{\times 4}}\right]$

Repeat Above Q for Rz Pulse





$H(t) = S_1(T_b\text{-}t) - S_2(T_b\text{-}t) = A \qquad 0 \le t \le T_b$



$P_e = Q\left[\sqrt{\frac{E_d}{2N_o}}\right]$

$E_d = \int_o^{T_b} |s_2(t) - s_2(t)|^2 \, dt = A^2T_b$

$P_e = Q\left[\sqrt{\frac{A^2T_b}{2N_o}}\right]$

$(E_b)_{avg} = P_oE_{bo} + P_1Eb_1$

$E_b = \frac{1}{2} \times o + \frac{1}{2} \times A^2T_b$

$P_e = Q\left[\sqrt{\frac{E_b}{N_o}}\right]$

$h(t) = S_1(T_b\text{-}t) - S_2(T_b\text{-}t)$

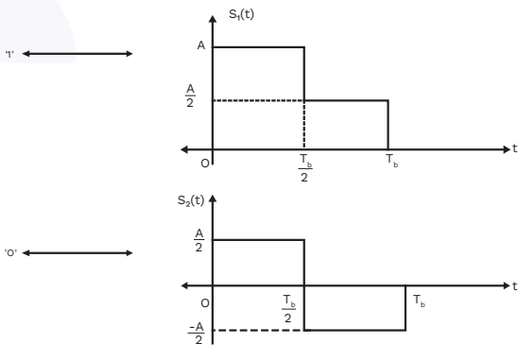
$h(t) = m(t) - n(t)$

$E_d = \int_o^{T_b} |S_1(t) - S_2(t)|^2 \, dt$

$= \int_o^{T_b} |S_2(t)|^2 \, dt + \int_o^{T_b} |s_2(t)|^2 - 2\int_o^{T_b} S_1(t)S_2(t)dt$

$E_d = E_{s_1} + E_{s_2} - 2\int_o^{T_b} S_1(t)S_2(t)dt$

Repeat above Q

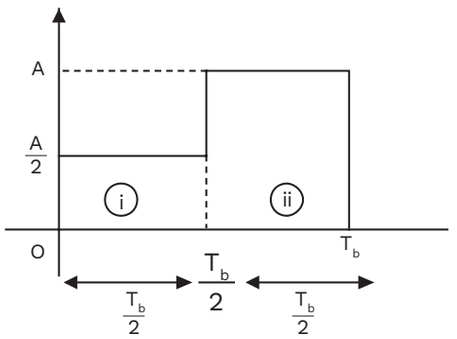


1. $H(t) = S_1(T_b\text{-}t) - S_2(T_b\text{-}t)$

2. $P_e = Q\left[\sqrt{\frac{E_d}{2N_o}}\right]$

$E_d = \int_o^{T_b} |P(t)|^2 dt$

$$P(t) = S_1(t) - S_2(t)$$

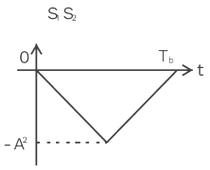


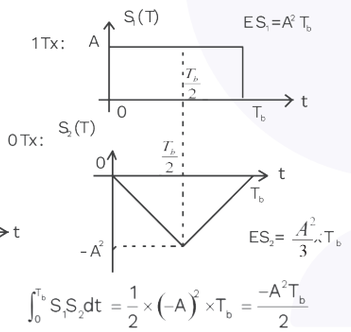
$$E_d = \left(\frac{A}{2}\right)^2 \times \frac{T_b}{2} + A^2 \times \frac{T_b}{2} = \frac{5A^2T_b}{8}$$

$$P_e = Q\left[\sqrt{\frac{5A^2T_b}{16N_o}}\right]$$

(iv)

Calculate P_e





$$P_e = Q\left[\sqrt{\frac{E_d}{2N_o}}\right]$$

$$E_d = \int_o^{Tb} |p^2(t) = E_{s1} + E_{s2} - 2\int_o^{Tb} S_1(t)S_2(t)dt$$

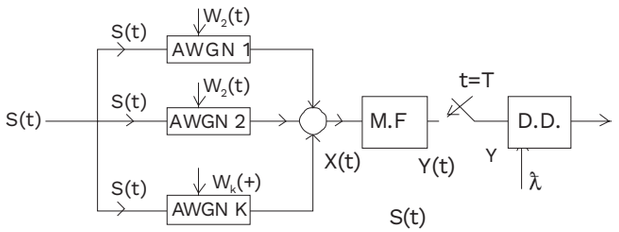
$$= A^2T_b + \frac{A^2}{3}T_b - 2\left\{\frac{-A^2T_b}{2}\right\}$$

$$Ed = \frac{7A^2T_b}{3}$$

$$Pe = Q\left[\sqrt{\frac{\sqrt{7A^2T_b}}{6N_o}}\right]$$

Transmission of Signal via Independent

Identically distributed AWGN channel



$$S(t) = \begin{cases} S_1(t): o \le t \le Tb: 1\ T_x \\ S_2(t): o \le t \le Tb: 0\ T_x \end{cases}$$

$$X(t) = Ks(t) + W_1(t) + w_2(t) + \cdots + W_k(t)$$

$$= \begin{cases} Ks_1(t) + W_1(t) + W_2(t) + \cdots + W_k(t): 1Tx \\ Ks_2(t) + W_2(t) + W_2(t) + \cdots + W_k(t): 0Tx \end{cases}$$

$$W_i(t)\ White\ Noise + WSSP$$
$$E[W_i(t)] = o \quad i = 1to\ K$$
$$S_w : (f) = \frac{N_o}{2}$$

$$Y(t) = \begin{cases} Ka_1(t) + N_1(t) + N_2(t) + \ldots\ldots\ldots + N_k(t) & 1T_y \\ Ka_2(t) + N_1(t) + N_2(t) + \ldots\ldots\ldots + N_k(t) & oT_y \end{cases}$$

$$Ni(t) \rightarrow \text{zeromean GR WSSRP}$$

$$Y = \begin{cases} Ka_1 + N_1 + N_2(t) + \ldots\ldots\ldots + N_k & 1T_x \\ \sim\sim\sim\

$$P\left(\frac{1}{oT_y}\right) = Q\left[\frac{\lambda-\mu_2}{\sigma_2}\right] = Q\left[\frac{\lambda-Ka_2}{\sigma_y}\right]$$

$$P\left(\frac{o}{1T_x}\right) = 1-Q\left[\frac{\lambda-\mu_1}{\sigma_1}\right] = 1-Q\left[\frac{\lambda-Ka_1}{\sigma_y}\right]$$

Since $\sigma^2_1 = \sigma^2_2$ and if $P(oT_x) = P(1T_x) = 1/2$

$$\lambda_{opt} = \frac{\mu_1+\mu_2}{2} = \frac{K(a_1+a_2)}{2}$$

$$P_e|_{min} = Q\left[\frac{\mu_1-\mu_2}{2\sigma_y}\right] = Q\left[\frac{K(a_1-a_2}{2\sigma_y}\right]$$

$$= Q\left[\frac{K(a_1-a_2}{2\sqrt{K}\sigma_N}\right]$$

$$\boxed{P_e|_{min} = Q\left[\sqrt{K}\left(\frac{a_1-a_2}{2\sigma_N}\right)\right]}$$

Conclusion 1) $\boxed{P(0T_x)P(1T_x) = \frac{1}{2}$ 1AWGN Channel $\lambda_{opt}}$

$\boxed{\sigma_Y = \sigma_N}$

$$\boxed{P_e|_{min} = Q\left[\frac{\mu_1-\mu_2}{2\sigma_Y}\right]}$$

Matched Filter

$$\boxed{P_e|_{min}|_{min} = Q\left[\sqrt{\frac{E_d}{2N_o}}\right]}$$

$\boxed{P(oT_x) = P(1T_x) = \frac{1}{2}$ K AWGN Channel $\lambda_{opt}}$

$\boxed{\sigma^2_N$: Vanance of anyone R.V. $\{N_i\}}$

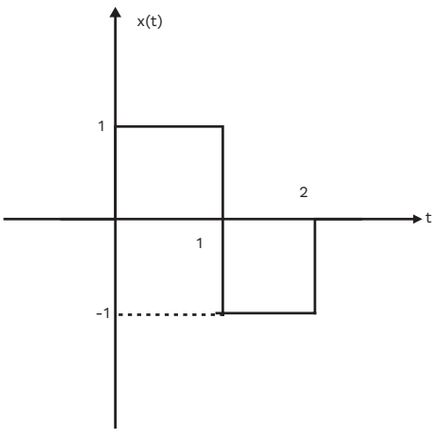
$$\boxed{P_e|_{min} = Q\left[\sqrt{K}\left(\frac{\mu_1-\mu_2}{2\sigma_N}\right)\right]}$$

Matched filter

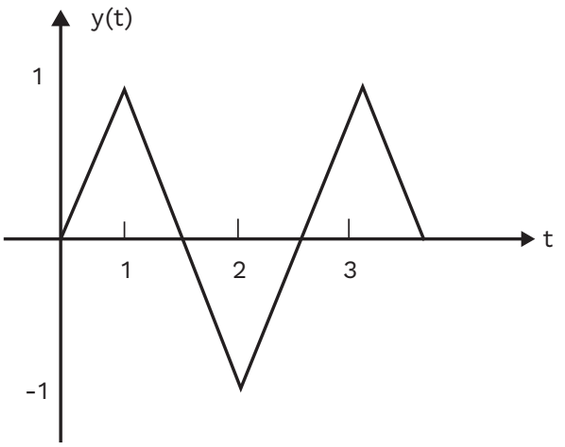
$$\boxed{P_e|_{min}|_{min} = Q\left[\sqrt{K}\sqrt{\frac{E_d}{2N_o}}\right] = Q\left[\sqrt{\frac{E_d}{2N_o}}\right]}$$

Exercise

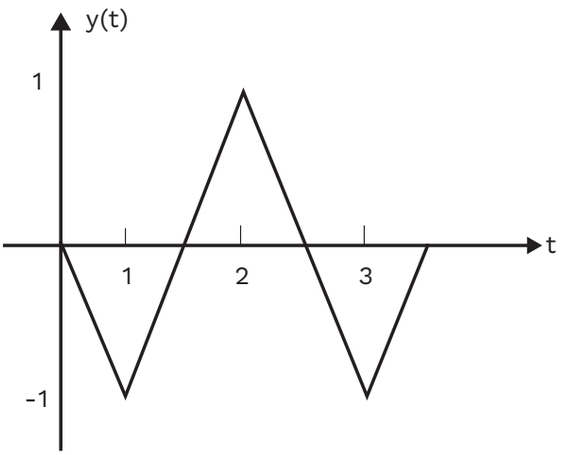
1. A signal as shown in the figure is applied to a matched filter. Which of the following does represent the output of this matched filter?



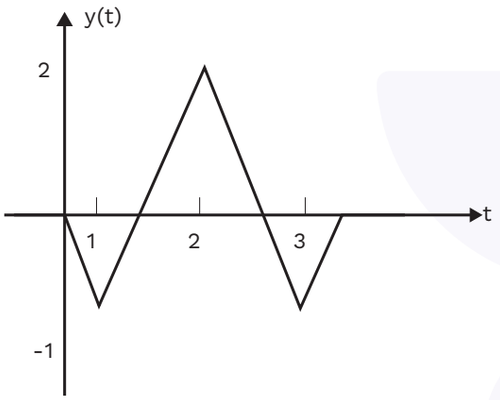
(A)



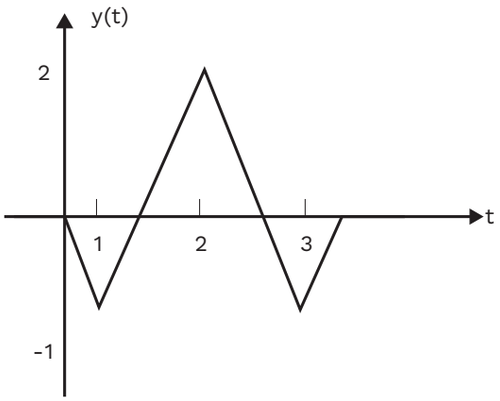
(B)



(C)



(D)



Common data for questions 2 and 3:

Let, g(t) = p(t) ⊗ p(t) where ⊗ does not convolution and

p(t) = u(t)-u(t-1) with u(t) being the step function.

2. The impulse response of filter matched to the signal
s(t) = g(t)-S(t-2) ⊗ g(t) is given as:

(A) s(1-t)

(B) –s(1-t)

(C) –s(t)

(D) s(t)

3. An amplitude modulation signal is given as X_{AM} = 100
[p(t)+0.5g(t)] $\cos_0 t$ in the interval $0 \le t \le 1$. One set of possible values of the modulation signal and modulation index would be?

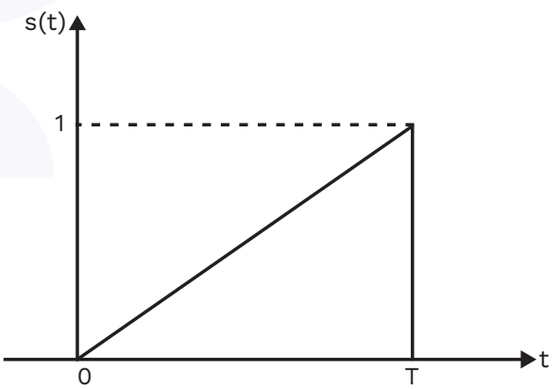
(A) t, 0.5

(B) t, 1.0

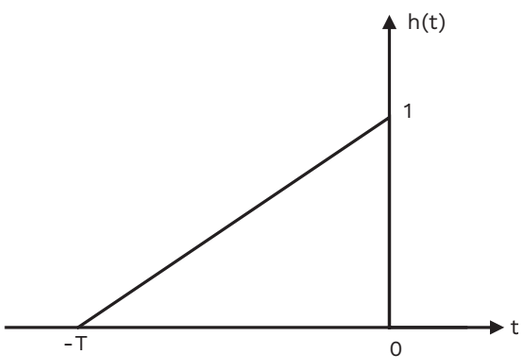
(C) t, 2.0

(D) t^2, 0.5

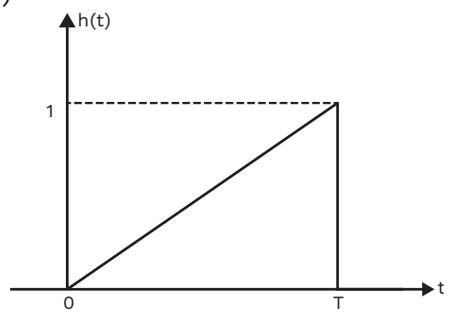
4. Consider the pulse shape s(t) as shown. The impulse response h(t) of the filter matched to this pulse is:



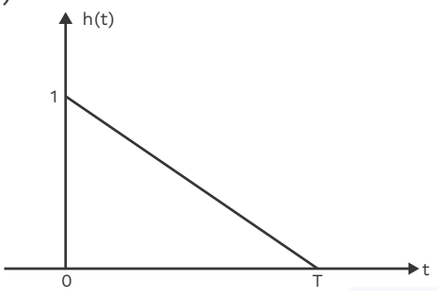
(A)



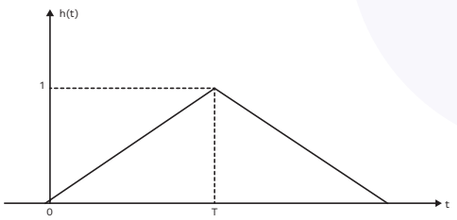
(B)



(C)

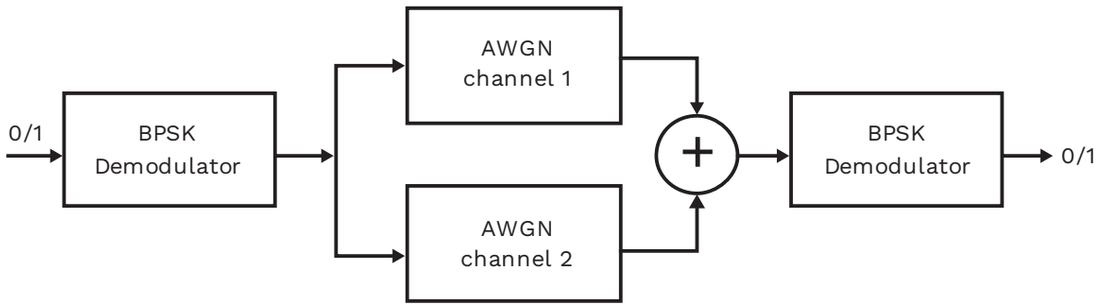


(D)



5. Let, $Q\left(\sqrt{\gamma}\right)$ be the BER of a BPSK system over an AWGN channel with two-side noise power spectral density $N_o/2$. The parameter γ is a function of bit energy and noise power spectral density.

A system with two independent and identical AWGN channels with noise power spectral density $N_o/2$ is shown in figure. The BPSK demodulator receives the sum of output of both the channels.



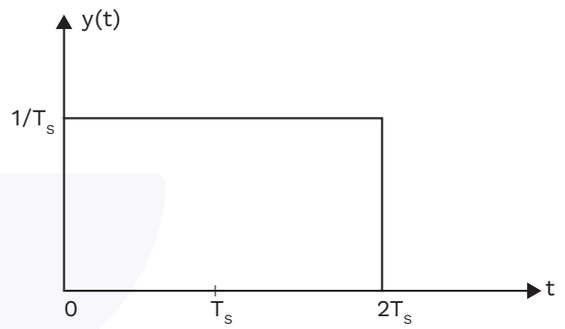
If the BER of this system is $Q\left(b\sqrt{\gamma}\right)$, then the value of B is _____.

6. A binary baseband digital communication system employs the signal

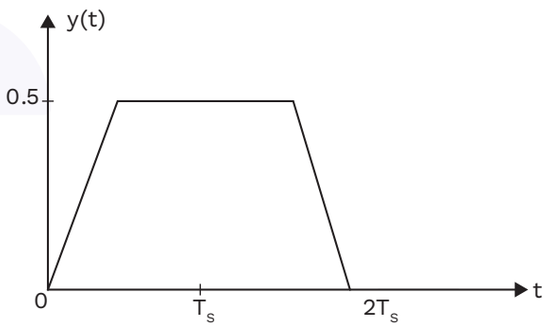
$$p(t)=\begin{cases}\dfrac{1}{\sqrt{T_s}} & 0 \le t \le T_s \\ 0, & otherwise\end{cases}$$

For transmission of bits, the graphical representation of the matched filter output y(t) for this signal will be

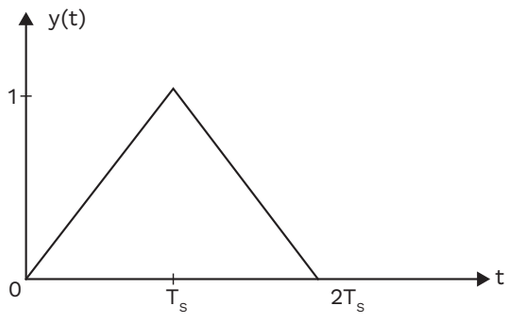
(A)



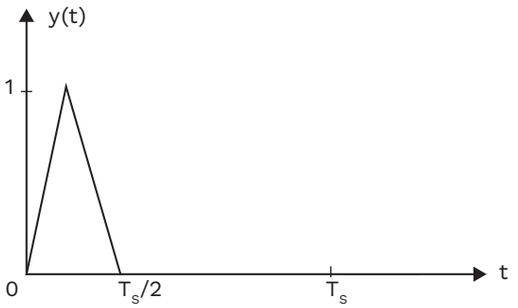
(B)



(C)

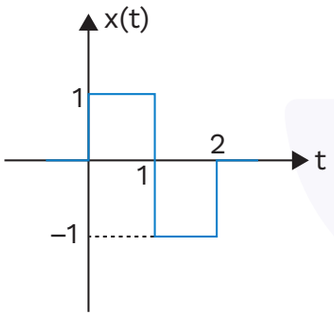


(D)



Solutions

1.



From the above figure, x(t) can be written as,

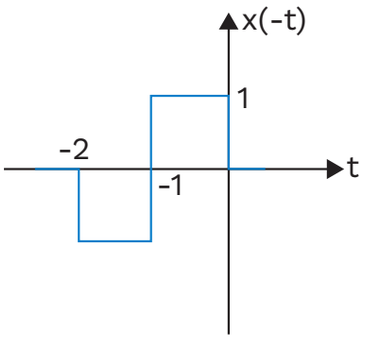
$$x(t) = u(t) - 2u(t-1) + u(t-2)$$

$$h(t) = x(T-t)$$

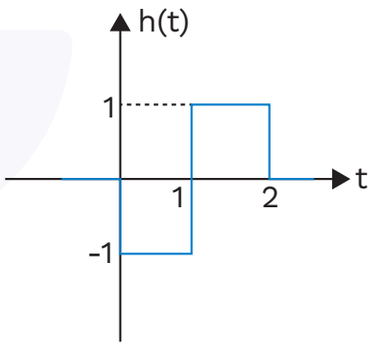
Here, T = 2

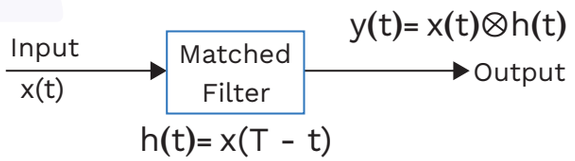
$$h(t) = x(2-t)$$

First draw x(-t) considering T = 0 i.e., folding operation on x(t).



After folding the signal, if we advance the signal it actually get delayed. The impulse response of matched filter is shown in below figure.





$$h(t) = -u(t) + 2u(t-1) - u(t-2)$$

Output of matched filter is given by,

$$y(t) = x(t) \otimes h(t)$$

Property of convolution,

$$\frac{d}{dt}y(t) = x(t) \otimes \frac{d}{dt}h(t)$$

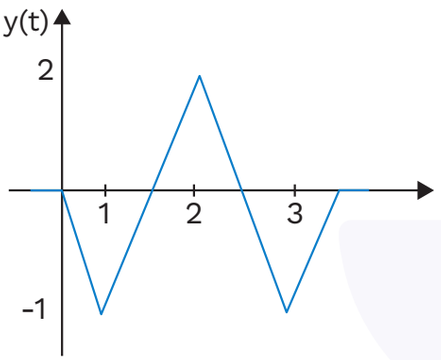
$$\frac{d}{dt}y(t) = \{u(t) - 2u(t-1) + u(t-2)\} \otimes \{-\delta(t) + 2\delta(t-1) - \delta(t-2)\}$$

$$\frac{d}{dt}y(t) = -u(t) + 2u(t-1) - u(t-2) + 2u(t-1) - 4u(t-2) + 2u(t-3)$$

$$-u(t-2) + 2u(t-3) - u(t-4)$$

$$\frac{d}{dt}y(t) = -u(t) + 4u(t-1) - 6u(t-2) + 4u(t-3) - u(t-4)$$

$$\therefore y(t) = -r(t) + 4r(t-1) - 6r(t-2) + 4r(t-3) - r(t-4)$$



Hence, the correct option is (c).

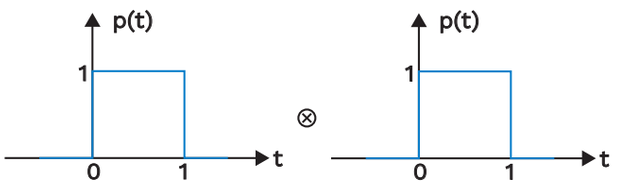
2.

Given: $g(t) = p(t) \otimes p(t)$ *and*

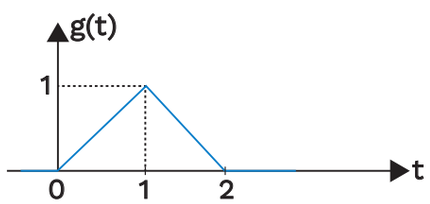
$$p(t) = u(t) - u(t-1)$$

$$s(t) = g(t) - \delta(t-2) \otimes g(t)$$

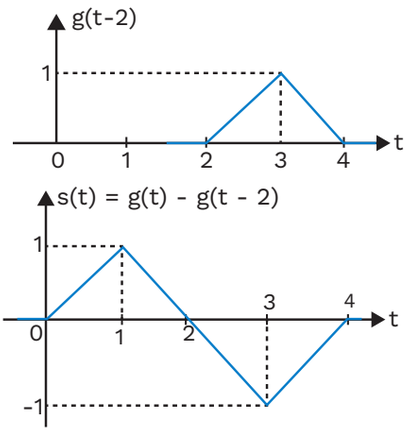
$$s(t) = g(t) - g(t-2)$$



Convolution of two symmetrical rectangular pulses is triangular signal.



After delaying the signal g(t) by 2 sec,



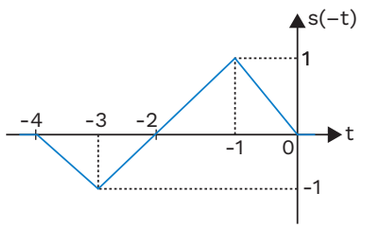
The impulse response of matched filter is given by,

$$h(t) = s(T-t)$$

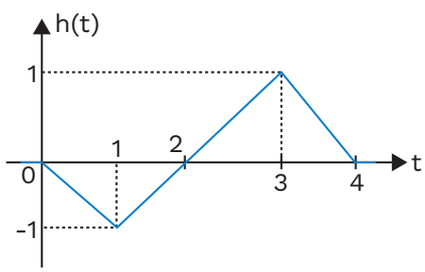
Here, T = 4

$$h(t) = s(4-t)$$

First draw s(-t) considering T = 0 i.e., folding operation on s(t).



After folding the signal, if we advance the signal it actually get delayed. The impulse response of matched filter is shown in below figure.

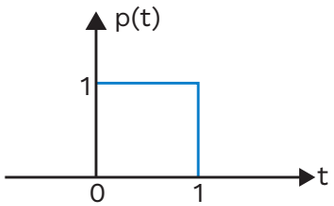


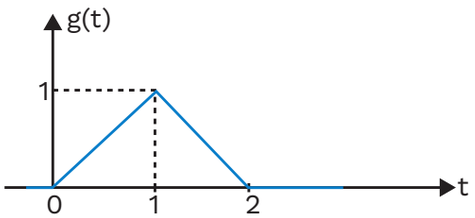
From figure it can be seen that,

$$h(t) = -s(t)$$

Hence, the correct option is (c).

3.





$$p(t) = 1, \qquad 0 < t < 1$$

$$g(t) = \begin{cases} t, & 0 < t < 1 \\ 1 - t, & 1 < t < 2 \end{cases}$$

So, $X_{AM}(t) = 100\,[1 + 0.5t] \cos \omega_0 t$ ...(i)

The expression of AM signal is given by,

$$X_{AM}(t) = A_c\left[1 + k_a m(t)\right]\cos \omega_0 t \qquad \text{...(ii)}$$

On comparing equations (i) and (ii),

$$k_a = 0.5, \quad m(t) = t$$

In AM, modulation index is given by,

$$m_a = k_a \left|m(t)\right|_{max} = 0.5 \times 1 = 0.5$$

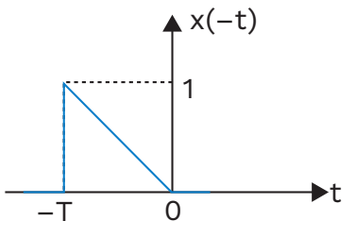
Hence, the correct option is (a).

4.

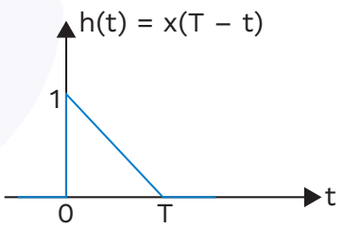
The response of the matched filter given by,

$$h(t) = x(T - t)$$

First draw x(–t) considering T = 0 i.e., folding operation on x(t).



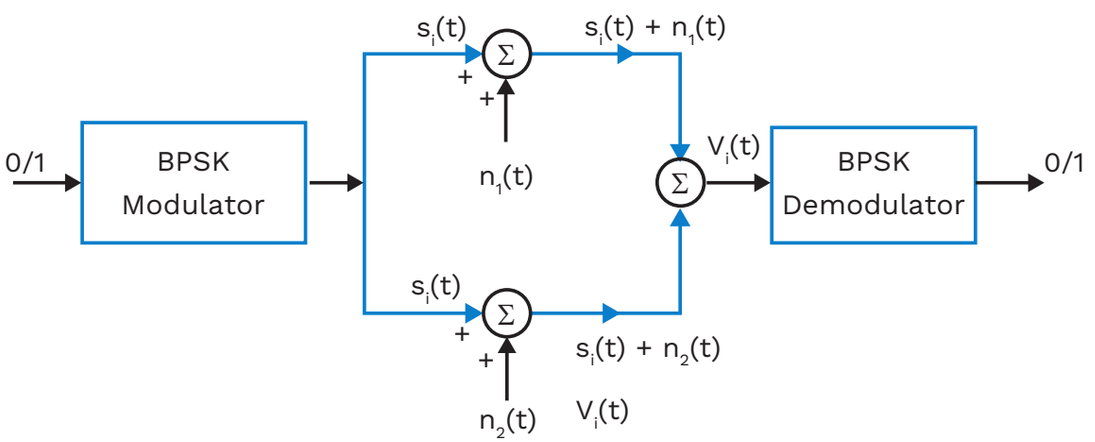
After folding the signal, if we advance the signal it actually get delayed. The impulse response of matched filter is shown in below figure.



Hence, the correction is (c).

5.

A system with two independent and identical AWGN channels with noise power spectral density $N_0 / 2$ is shown in the figure.



In BPSK, bit error rate or probability of error is given by,

$$BER\left(Bit\ Error\ Rate\right) = Q\left(\sqrt{\frac{2E_s}{N_0}}\right)$$

$$BER = Q\left(\sqrt{\frac{E_s}{N_0/2}}\right) = Q\left(\sqrt{\frac{Energy\ of\ signal}{Noise\ PSD}}\right)$$

$$BER = Q\sqrt{\gamma}$$

$$V_i\left(t\right) = 2s_i\left(t\right) + n_1\left(t\right) + n_2\left(t\right)$$

Signal energy at demodulator input can be calculated as,

$$E'\left(new\ energy\right) = \int_{-\infty}^{\infty}\left|2s_i\left(t\right)\right|^2 dt$$

$$E'\left(new\ energy\right) = 4\int_{-\infty}^{\infty} s_i^2\left(t\right)dt = 4E_s$$

Signal energy gets four fold after passing through the given system.

$$N' = PSD\left[n_1\left(t\right) + n_2\left(t\right)\right]$$

$$N' = \frac{N_0}{2} + \frac{N_0}{2} = N_0$$

$$\text{New BER} = \text{BER}' = Q\sqrt{\frac{E'}{N'}} = Q\sqrt{\frac{4E_s}{N_0}}$$

$$BER' = Q\sqrt{2\left(\frac{E_s}{N_0/2}\right)}$$

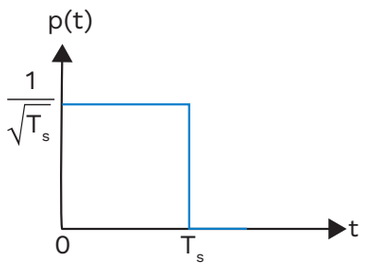
$$BER' = Q\sqrt{2\gamma} = Q\left(\sqrt{2}\times\sqrt{\gamma}\right)$$

$$b = \sqrt{2} = 1.414$$

Hence, the value of b is 1.414.

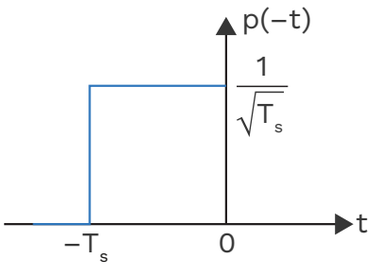
6. Solution:

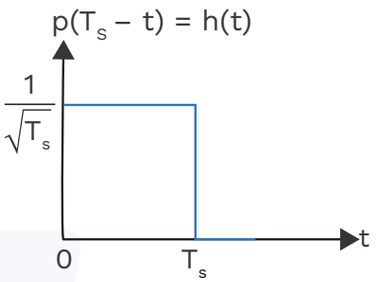
Given: $p\left(t\right) = \begin{cases} \frac{1}{\sqrt{T_s}} & 0 \le t \le T_s \\ 0 & otherwise \end{cases}$



Impulse response of matched filter

$$h\left(t\right) = p\left(T_s - t\right)$$





After folding the signal if we advance the signal it actually gets delayed.

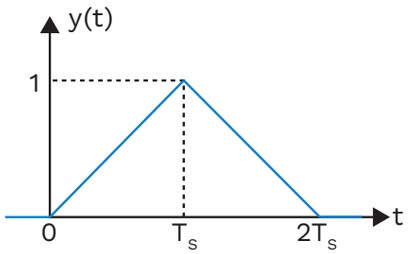
Output of matched filter

$$y\left(t\right) = p\left(t\right)\otimes h\left(t\right)$$

Note: Convolution of two rectangular signals of same duration results in triangular signal.

Amplitude of output signal = Energy of the input signal

$$= \left(\frac{1}{\sqrt{T_s}}\right)^2 \times T_s = 1$$



Hence, the correction option is (c).